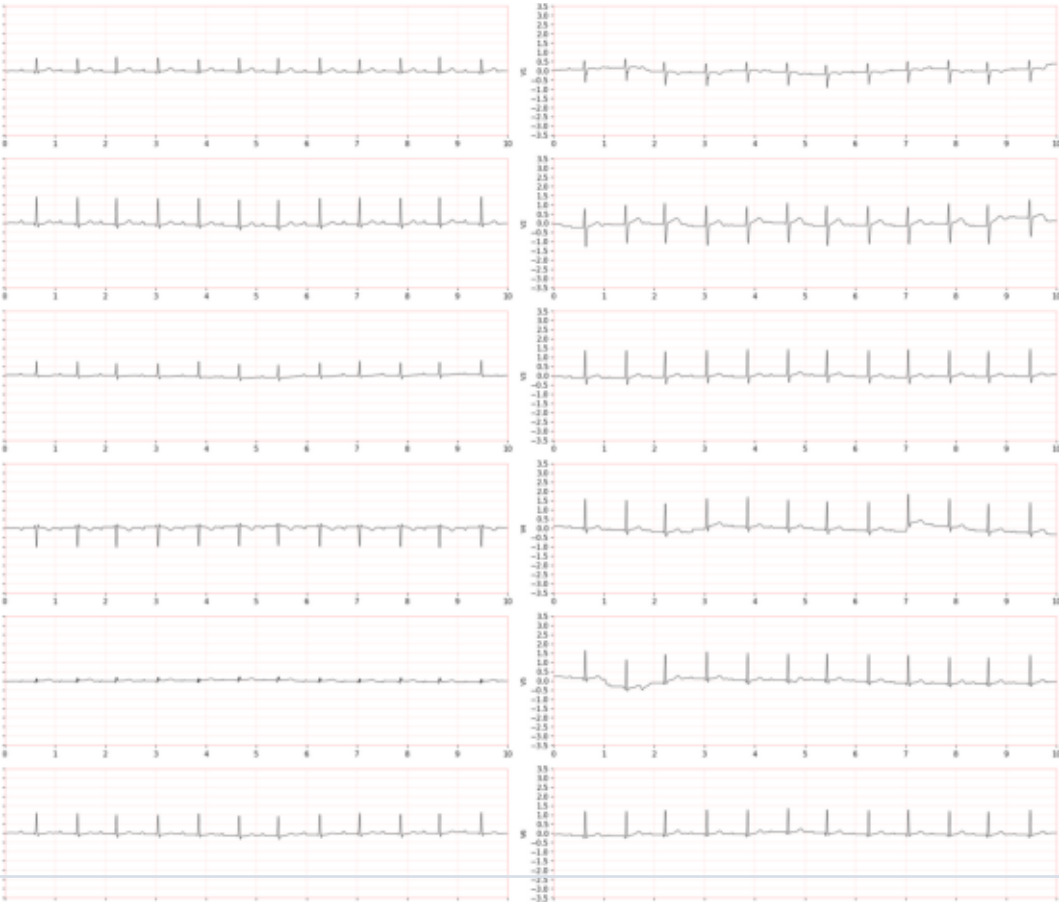
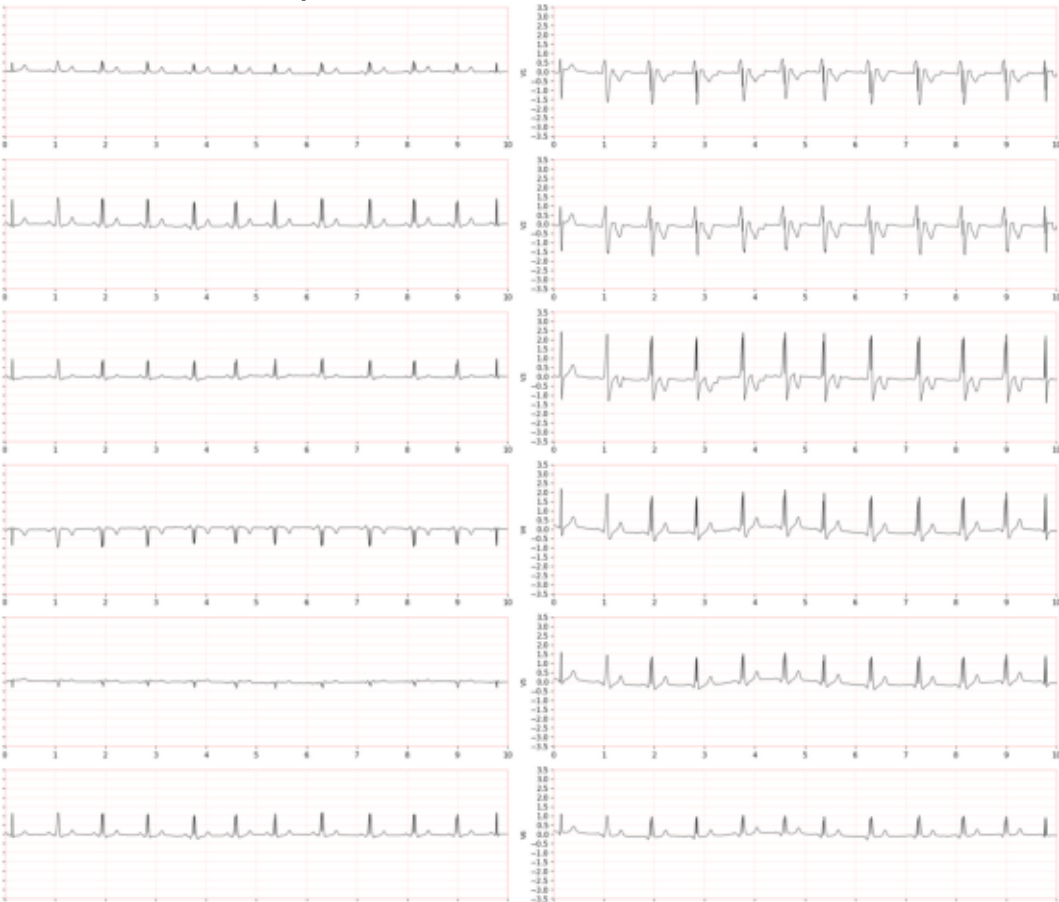


Exact historical model-input ECG



Exact historical model-input ECG



Case 1 · Existing complex five-option MCQ

First-degree AV block: integrated interpretation

wf_08_00439 · competency: integrated_interpretation

Treat the image as the complete case evidence and distinguish direct observations from inferred mechanism. Which synthesis of the major ECG findings is internally consistent with the tracing? Options:

- ✓ A. First-degree atrioventricular block: Every P wave conducts to a QRS, but the PR interval is uniformly prolonged.
- B. Second-degree av block, mobitz i (wenckebach): Conducted PR intervals progressively lengthen until a P wave is not followed by a QRS.
- C. Second-degree av block, mobitz ii: Conducted beats have essentially constant PR intervals, followed by an abrupt nonconducted P wave.
- D. Sinus bradycardia: The rhythm is regular and sinus in origin but has a rate below 60 beats/min.
- E. Third-degree (complete) atrioventricular block: P waves and QRS complexes march independently with no stable PR relationship.

Qwen3.8 thinking CoT: PASS · boxed A · 26,441 characters

Full CoT is retained in selected_complex_cases_2.jsonl; the figure shows the unmodified model-input question.

Historical closed-loop waveform source; automated label, not clinician-adjudicated

Case 2 · Existing complex five-option MCQ

RBBB: mechanism localization

wf_13_00434 · competency: mechanism_localization

Treat the image as the complete case evidence and distinguish direct observations from inferred mechanism. Which site of impulse formation, conduction delay, or myocardial injury is most consistent with the tracing? Options:

- A. For left bundle branch block, block in the left bundle forces the left ventricle to depolarize late through slower myocardial spread. Atrial impulses may conduct 1:1, but ventricular depolarization spreads abnormally after right-bundle activation.
- B. For right ventricular hypertrophy with strain pattern, increased right-ventricular muscle mass shifts depolarization forces anteriorly and rightward. The QRS vector shifts toward the right ventricle while secondary repolarization opposes the dominant rightward force.
- ✓ C. For right bundle branch block, block in the right bundle delays right-ventricular depolarization after normal left-sided activation. Atrial impulses can conduct 1:1 while left-ventricular activation precedes delayed right-ventricular activation.
- D. For a premature ventricular complex, an ectopic ventricular focus depolarizes myocardium before the next sinus-conducted beat. The premature ventricular beat is dissociated from the expected sinus P-to-QRS sequence and is followed by a compensatory pause.
- E. For ventricular pre-excitation (Wolff-Parkinson-White pattern), an accessory atrioventricular pathway bypasses the normal AV-nodal delay and activates ventricular myocardium early. Part of the ventricle activates before the normal AV-His-Purkinje impulse completes conduction.

Qwen3.8 thinking CoT: PASS · boxed C · 19,614 characters

Full CoT is retained in selected_complex_cases_2.jsonl; the figure shows the unmodified model-input question.

Historical closed-loop waveform source; automated label, not clinician-adjudicated